

PII Shield AI – PII Detector & Masker

1. Purpose of AI Usage

Artificial Intelligence tools were used as a development assistant to speed up the design and implementation of this project.

AI helped in understanding concepts, generating boilerplate code, improving structure, and assisting with documentation. However, all final decisions, debugging, and integration were done manually by the developer.

2. What AI Helped With

✓ Code Development

Flask backend structure

CSV processing logic

Regex-based PII detection (Email, Phone, PAN, Aadhaar)

Data masking functions

Report generation logic

✓ Debugging Support

Fixing runtime errors

Resolving Pandas data type issues

Fixing Flask route errors

Improving test case structure

✓ Frontend Design

Dashboard layout design

Analytics UI (charts, counters, compliance score)

CSS styling improvements

✓ Documentation

README.md generation

Project explanation writing

Architecture description

3. What AI Got Wrong

Some AI-generated outputs required correction:

Incorrect Pandas assignments causing dtype errors

Missing module imports in project structure

Over-simplified test cases not matching real logic

Minor Flask routing issues

UI layout required manual adjustments for proper alignment

All issues were fixed manually after testing.

4. Best Prompts Used

“Create a Flask PII detector that reads CSV and masks sensitive data”

“Write regex for email, phone, PAN, Aadhaar detection”

“Build a dashboard UI for analytics using HTML/CSS”

“Generate pytest test cases for Flask application”

“Fix Pandas dtype error while masking values in DataFrame”

5. Human Contribution

Final architecture design

Code integration and debugging

UI customization and styling

Testing and validation

GitHub repository management

Final submission preparation

6. Conclusion

AI was used only as an assistant tool to accelerate development. The final application is fully tested, manually verified, and structured by the developer.

 Prepared By

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